

FahMai: A Thai Grounded Tool-Use Benchmark for the Deployable Model Tier

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Abstract

We present **FahMai**, a bilingual (Thai/English) benchmark for *grounded tool-use*. The task is to answer employee-directory questions over a structured corporate database by issuing tool calls, or to refuse when the answer is not in the data. We design the questions to be inspired by a real Thai enterprise-directory deployment, while every record — names, nicknames, phone extensions, emails, and the org chart — is synthetic. The benchmark therefore carries no personal data and is contamination-free by construction. FahMai contains 626 items across 8 question groups (29 subtypes), and it targets phenomena specific to Thai directory search: nickname–formal-name resolution, idiosyncratic romanization, homonymous shared given-names, and a five-way refusal taxonomy. We grade every item deterministically, using substring and exact-count matching against unique-by-construction gold plus a fixed set of canonical refusal phrases, with no LLM judge. Our central design choice is to treat tool availability as an experimental variable: we run each item under four tool regimes (grep-only, structured search, both, and a Python REPL), which lets us measure not only *whether* a model answers correctly but *how* it uses tools. We find that the task is near-solved at the frontier (gpt-5.4 97–99%, GLM-5.1 95–98%, Sonnet 4.6 92–95%), yet remains sharply discriminating in the affordable, Thai-capable tier one would actually deploy (OpenThaiGPT-8B 22%, Typhoon-2.5-30B 57–73%). The tool axis also surfaces behavioral differences that a single accuracy number hides, including retry-after-failure, tool routing, and a frontier regression: gpt-5.5 collapses on the wide-schema search tool (74%) through a parameter-over-specification loop, while remaining at 96–97% on grep and the REPL. We argue that an easy, faithful, and airtight benchmark can be valuable for precisely these reasons, and we release FahMai as a deployment-relevant probe of Thai-language robustness in tool use.

1 Introduction

Most agentic benchmarks chase difficulty. They assemble tasks hard enough that even the strongest models fail often, on the premise that headroom is what makes a benchmark useful. In this work, we take the opposite starting point. The task we study is looking up a colleague’s phone extension, email, or manager in a company directory by calling a tool, and this task is easy. A capable model with a working search tool should solve it almost every time, and we show that the best current models essentially do. We argue, however, that “easy” hides a great deal, and we organize the paper around four reasons such a benchmark is still worth building and releasing.

(1) The difficulty is an empirical claim that, for Thai, has not been measured. It is widely assumed that grounded directory lookup is trivial for modern LLMs, but this assumption rests on intuitions formed in English. Thai directory search is not the same task. Thai is written without word spaces; people are addressed by nicknames (ชื่อเล่น) that bear no systematic relationship to their formal names; the same given name recurs across dozens of employees; romanizations are idiosyncratic and inconsistent; and honorific particles wrap names in ways that defeat naive string matching. For instance, whether a model can resolve “เบอร์รี่ของพี่อยู่กที่ SF” — *Hook’s number at the SF branch* — into the right row, or correctly *refuse* when the named person does not exist, is a question that had not been answered with an airtight grader. The benchmarks that might have answered it are almost all English (Patil et al., 2024; Qin et al., 2024; Yao et al., 2024), and Thai evaluation suites target language understanding rather than grounded tool-use (Pipatanakul et al., 2023; Susanto et al., 2025). Turning a shared assumption into a measurement is therefore a contribution in its own right, and, as we show in point (4),

the assumption turns out to be only partly correct.

(2) The benchmark is grounded in a real task.

We design FahMai’s questions to be inspired by a real Thai enterprise-directory assistant, that is, the kinds of requests people actually make of such a system. The underlying data is fully synthetic (Section 3.2), which is what lets us release the benchmark without exposing any information and what makes it contamination-free. What we carry over from the real setting is only the *shape* of the task: the mix of nickname lookups, branch-scoped queries, counterfactual premises, code-switching, and out-of-scope requests, not any content. The result, we believe, is a query distribution that reflects how people query a Thai company directory, rather than what a benchmark author imagines they would.

(3) The discriminating power is concentrated in the tier one would actually deploy.

In practice, no one runs a frontier reasoning model to fetch an extension; the economically relevant systems are 8B–30B models, ideally Thai-specialized ones, where latency and cost are tolerable at directory-query volume. It is exactly there that FahMai spreads models out. For instance, a small Thai-native model (OpenThaiGPT-8B) sits at 22% and is effectively tool-blind, verbalizing “I should search” without ever emitting a tool call, while a mid-size general model clears 95%. The benchmark thus speaks to a concrete procurement question: which affordable, Thai-capable model can one trust to ground answers over one’s own directory? The frontier’s role here is not to be challenged but to certify the ceiling. When the strongest models saturate an airtight benchmark, the items are demonstrably well-posed and solvable, which suggests that the large gaps below reflect genuine capability differences rather than benchmark artifacts.

(4) Even at the frontier, “easy” hides surprises, and they live in *how* tools are used rather than in the final score.

Our core methodological move is to treat tool availability as an experimental variable (Section 3.4). We evaluate every item under four tool regimes: a raw-text grep, a structured search with many optional fields, both together, and a Python REPL. By holding the item fixed and varying only the toolset, we expose process-level behavior that a single accuracy number would erase. Three findings illustrate the point. First, retry behavior separates models: on noisy Thai names un-

der grep-only, gpt-5.4 re-queries after an empty result and recovers, whereas Typhoon greps the full noisy string once and never retries (0% on the same items). Second, tool routing matters: models with an always-search policy may pick the wrong tool when both are offered, which produces a roughly 40-point swing on identical items depending only on which tool wins. Finally, we observe a frontier regression. gpt-5.5 collapses to 74% on the wide-schema search tool by over-specifying its ~15 optional parameters with hallucinated values, receiving empty results, and looping to the round cap, yet the same model scores 96–97% with grep or the REPL. A newer model thus regressed on an easy task, in a way that only a multi-tool-config harness could localize. On a hard benchmark, “chose the wrong tool” is indistinguishable from “task too hard”; on an easy and airtight one, the mechanism becomes legible.

Taken together, these points suggest that a benchmark need not be hard for the best model to be worth releasing. It needs instead to be faithful to a real task, airtight enough that a number means something, and discriminating for the models one would actually deploy. When accuracy saturates at the top, the informative axes shift to which cheaper model still works, how robustly it handles Thai, how it behaves across tool regimes, and how efficiently it answers in cost and latency, which we also report. We design FahMai around those axes. In this work, we contribute the following:

- **FahMai**, a 626-item bilingual Thai/English benchmark for grounded directory tool-use, fully synthetic (no PII, contamination-free) yet grounded in a real query distribution, with airtight deterministic grading and no LLM judge.
- A **29-subtype question taxonomy** covering Thai-specific retrieval phenomena (nickname/romanization resolution, homonym disambiguation, multi-hop org bridges) and a principled **five-way refusal taxonomy**, each refusal reason bound to its own canonical phrase.
- A **tool-availability experimental design** (four regimes per item) that measures tool-use *process* — retry, routing, schema-induced failure — not only outcome.
- An evaluation across the frontier and the deployable tier showing the task is near-solved

at the top and sharply discriminating below, including a documented **gpt-5.5 tool-schema regression**.

2 Related Work

Tool-use and function-calling benchmarks. A line of work evaluates whether LLMs can invoke external tools and APIs correctly. Gorilla connects an LLM to large API collections, targeting hallucination-free call generation, and seeds the Berkeley Function-Calling Leaderboard (Patil et al., 2024). ToolLLM/ToolBench scales this to 16,000+ real-world REST APIs with automatically constructed instructions and solution paths (Qin et al., 2024). API-Bank supplies 73 runnable tools and 314 annotated dialogues to test planning, retrieval, and invocation (Li et al., 2023b). τ -bench moves toward interactive settings, emulating user-agent conversations under domain policies and reporting pass^k reliability, on which even strong function-calling agents succeed on under half of tasks (Yao et al., 2024). Broader agent benchmarks embed models in richer environments — AgentBench across eight interactive settings (Liu et al., 2024), WebArena in reproducible live websites (Zhou et al., 2024), and AppWorld in a controllable world of 457 APIs graded by state-based unit tests (Trivedi et al., 2024) — and deliberately pursue difficulty and long-horizon complexity, on which even GPT-4-class agents complete only a fraction of tasks. Across this literature, however, evaluation is almost exclusively **English** and uses a **single, fixed toolset**: tool availability is a property of the harness, not a variable under study. FahMai differs on three axes — it is Thai/English, grades against an airtight deterministic key, and makes the toolset itself an experimental variable so that tool-use *process* (retry, routing, schema-induced failure) is measured directly — and, by design, targets an *easy* task to expose the deployable tier rather than chasing frontier-level difficulty.

Grounded query benchmarks. Text-to-SQL evaluates grounded querying over structured data: Spider tests cross-domain generalization to unseen schemas and queries (Yu et al., 2018), and BIRD scales to large, “dirty” real-world databases where execution accuracy remains far below human performance (Li et al., 2023a). These grade an executed query against a gold result but are not agentic — the model emits a single query rather than choosing and orchestrating tools over multiple rounds.

FahMai keeps the airtight, execution-style grading of this tradition while embedding it in an agentic, multi-tool loop.

Refusal and abstention. Knowing when *not* to answer is a long-standing evaluation target; SQuAD 2.0 pairs answerable questions with adversarial unanswerable ones so that systems must abstain rather than guess (Rajpurkar et al., 2018). FahMai operationalizes abstention in a tool-use setting with a five-way refusal taxonomy (field-absent, person-not-found, subjective, out-of-company, field-blank), each bound to a canonical phrase and a universal no-leak guard, so refusal correctness is graded as precisely as answer correctness. In the agentic setting this connects to tool-use safety: ToolEmu emulates tool execution to surface high-stakes agent failures such as leaking private data (Ruan et al., 2024); FahMai’s no-leak guard and out-of-company refusals are a lightweight, deterministic counterpart, checking that an agent declines rather than over-shares.

Thai and Southeast-Asian LLM evaluation. Thai-specific evaluation has centered on language understanding and knowledge: Typhoon introduces Thai LLMs together with the ThaiExam knowledge benchmark (Pipatanakul et al., 2023), and SEA-HELM provides a holistic, continually-maintained suite across five SEA languages spanning NLP classics, linguistics, culture, and safety (Susanto et al., 2025). None of these target **grounded agentic tool-use**. To our knowledge FahMai is the first Thai — and first SEA-language — benchmark for grounded tool-use, and the first to treat tool availability as a controlled variable rather than a fixed harness choice. We adopt SEA-HELM’s bootstrap confidence-interval reporting for scores (Section 4).

3 The FahMai Benchmark

3.1 Design principles

FahMai is built on three commitments. **Airtight grading without an LLM judge:** every item has a gold answer that is unique by construction, graded by substring match (must_contain_any_of) or exact count, and every refusal item is graded against a fixed canonical phrase. This makes scores exactly reproducible and removes judge-model variance. **Synthetic data, real task:** the directory is fictional (Section 3.2) but the query types are inspired by a real deployment. **Tool availability as a variable:** the same items are run under four tool regimes (Sec-

tion 3.4), so the benchmark measures tool-use behavior, not just task accuracy.

3.2 The FahMai directory (data)

The knowledge base is a single synthetic corporate directory — **FahMai Co.**, a fictional Thai retail/energy conglomerate — of **1,995 employee rows**. Every field is generated: Thai and romanized names, nicknames, phone extensions, emails, position titles, and a four-level org hierarchy (unit $\subset$ section $\subset$ department, plus brand subsidiaries and branches). No record corresponds to a real person; no real phone number, email, or name pairing appears. We verified zero overlap with common Thai NLP corpora (VISTEC, Wisesight), so a model cannot have memorized any answer — the benchmark is contamination-free *by construction*, not merely by recency. This synthetic design is also what makes release possible: the task is inspired by a real enterprise-directory assistant whose underlying data cannot itself be published.

The directory is engineered to support airtight items: name and code lookups resolve to exactly one row; nicknames are deliberately *shared* across employees to create homonym-disambiguation items; org units have single, well-defined heads where multi-hop bridges require them; and a controlled number of fields are left blank to support “field-present-but-empty” refusals.

3.3 Question types

The 626 items fall into **8 groups / 29 subtypes**, 38% English / 62% Thai, 521 answer items / 105 refusal items. The full subtype breakdown with examples is in Appendix A; the group structure is given in Table 1.

Several subtypes carry the benchmark’s intent and deserve emphasis:

- **Thai noisy-name resolution (B2, B3).** A query gives a nickname or a misspelled/honorific-wrapped name (“คุณกมล ชัยส...”, “Hook from SF”) that the model must normalize to a formal record. The *hard* tier gives only a shared given name and a company-unique role, forcing normalization **and** disambiguation among 9–13 homonyms.
- **Homonym grids (D1).** A bare nickname (“มีน คือใคร”) maps to several people; the gold is the full set, testing whether the model enumerates rather than guesses one.

grp	n	theme
A	65	Direct identity / canonical-code lookup — the understanding floor.
B	95	Nickname & noisy-name resolution (diminutives, honorifics, branch-scoped, homonyms).
C	99	Counting & aggregation (nickname counts, org-unit headcount, filtered counts, surname-family, superlative).
D	70	Disambiguation — pick the right person against a near-miss / negative constraint.
E	87	Multi-hop, bridge & hierarchy (section/dept-head bridges, secretary→exec, implicit hierarchy, deep 4-level chains).
F	65	Org & brand knowledge (subsidiary routing, premise-correction, in-house brand ops).
G	40	Bilingual / code-switch — the same fact asked across Thai↔English.
H	105	Refusal & safety — five distinct refusal reasons, each with its own canonical phrase.

Table 1: The 8 question groups.

- **Multi-hop org bridges (E1, E2, E5).** No explicit manager edge exists in the data, so depth comes from org-unit nesting and the secretary↔executive bridge — e.g. the extension of the secretary of the VP of the department that employee X belongs to (a 4-hop chain). Phrasing uses natural Thai relative-clause framing, never naming the intermediate people.
- **Counterfactual premise-correction (F2).** The query asserts a false premise (“I heard Kamala is the CTO, what’s her number”) and the model must correct it (she is the CFO) rather than comply.
- **Code-switch (G3).** Thai-framed questions about English-titled entities (“ขอ email ของ Chief Executive Officer”) and vice-versa.
- **Five-way refusal taxonomy (H).** Refusals are not monolithic: field-not-in-table (salary, age), person-not-found (plausible but absent combo-names), subjective (“who is the best engineer”), out-of-company (a competitor’s CTO), and field-present-but-blank (no nickname on record). Each reason has its own canonical phrase, and all H items additionally carry a universal “never leak an extension” guard.

config	tools available	probes
T1 grep-only	grep_csv (regex over raw CSV) + read_csv_rows	retry-after-empty; raw-text robustness
T2 search	search_employees (structured, ~15 optional fields)	structured-query formulation
T3 both	search + grep + read_rows	tool routing when alternatives compete
T4 repl	python_repl (pandas over the CSV)	frontier aggregation; REPL over-use

Table 2: The four tool configurations. Holding the item fixed and varying only the toolset isolates tool affordance.

3.4 Tools and the tool-availability axis

We expose three tool *types* and run every item under four tool configurations (Table 2). Because the only thing that changes across configs is the available toolset, differences in accuracy or behavior are attributable to tool affordance, not to the item. This is the design’s backbone and the source of the process-level findings in Section 5 (retry rate under T1, routing under T3, the gpt-5.5 schema loop under T2). We additionally derive metrics that older taxonomies treated as separate task types — e.g. retry behavior is read off T1, and tool-choice quality off T3 — rather than authoring them as their own items.

3.5 Grading

Answer items are graded by `must_contain_any_of` over a gold set that includes both Thai and romanized forms of the target (names, codes, emails), or by exact count for aggregation items. Refusal items are graded against the canonical phrase for their refusal reason, accepted in either language. There is **no LLM judge**: every verdict is a deterministic string operation, so the score is exactly reproducible. Gold answers are unique by construction and verified airtight (the gold substring does not match any non-target row). We report accuracy with SEA-HELM-style bootstrap confidence intervals, plus per-group and per-tool-config breakdowns, and — because the task is easy enough that efficiency is a real axis — per-model **cost and latency**.

4 Experimental Setup

Models. We evaluate **twelve model configurations** (eleven distinct models; gpt-5.5 at two

reasoning efforts) across three deployment tiers. The frontier tier comprises gpt-5.4 (medium), gpt-5.5 (medium and low), and Claude Sonnet 4.6. The mid-size open tier comprises GLM-5.1, DeepSeek-V4-Pro, DeepSeek-V4-Flash, Gemini-3-Flash, Gemma-4-31B, and MiniMax-M2.7. The Thai-specialized tier comprises Typhoon-2.5 (30B) and OpenThaiGPT-8B. We reach models through their native interfaces (the OpenAI Responses API for gpt-5.x, the Anthropic Messages API for Claude, the OpenTyphoon and Thai national gateways for the Thai-specialized models, Z.ai for GLM, and OpenRouter for the remaining open models), so each model uses its intended tool-calling format rather than a lowest-common-denominator shim.

Tool configurations. We run every item under the four tool regimes of Section 3.4. The frontier and free-to-run models (gpt-5.x, Sonnet, Typhoon, OpenThaiGPT) are evaluated under all four; the metered open models are run under T1/T2/T4, omitting T3 (“both”). The tool-routing signal that T3 isolates is read primarily off the always-available models, so dropping it for the metered set is a cost-control choice that leaves every per-tool comparison intact.

Inference. Decoding is greedy (temperature 0) where the provider supports it; reasoning-native models (gpt-5.x, DeepSeek-V4, GLM, MiniMax) use their default reasoning settings, and temperature is omitted for models that reject it. Each item is an agentic loop: the model receives a fixed system prompt and the question, issues tool calls, and is fed tool results until it emits a final answer or reaches a cap of **seven tool-call rounds**. Round-exhaustion counts as a failure — and is itself an informative signal, as it is precisely the mechanism behind the gpt-5.5 search collapse (Section 5). All runs write per-item JSONL traces and are resume-safe.

Knowledge base. Every item is graded against, and every tool operates over, a single frozen directory — `employees_v02.csv` (1,995 synthetic rows).

Scoring and confidence intervals. Accuracy is the fraction of items passing the airtight grader of Section 3.5. Because decoding is greedy, a model’s per-item output is stable across repeats, so we run a **single deterministic pass** per (model, config) cell and quantify uncertainty by **bootstrapping over items** (2,000 resamples, 95% percentile

CI), following SEA-HELM’s bootstrap-CI reporting. The eight-run stochastic averaging that SEA-HELM uses for sampled decoding addresses a variance source that greedy decoding removes. We report overall accuracy, per-group and per-subtype breakdowns, the per-tool-config comparison, and — because the task is efficiency-sensitive — per-model **cost** (USD from provider token pricing) and **latency** (per-item wall-clock; batched runs are flagged, since their round-trip timing is not per-item latency).

5 Results

5.1 Leaderboard

Table 3 gives per-config accuracy over all 626 items, with item-level bootstrap 95% CIs. Cells marked “—” are configs we did not run for the metered models (T3 dropped, Section 4).

The task is near-solved at the frontier and sharply discriminating below. Six models exceed 95% on their best config (gpt-5.4, gpt-5.5, GLM-5.1, DeepSeek-V4-Pro, DeepSeek-V4-Flash, Sonnet 4.6); the airtight grader and the frontier saturation together certify that the items are well-posed. Below the frontier the spread is wide and orderly: a capable-mid band at ~85–90% (Gemini-3-Flash, Gemma-4, MiniMax), then a steep drop to the Thai-specialized small tier — Typhoon-2.5 at 57–73% and OpenThaiGPT-8B at ~22%. The bootstrap CIs are tight enough (half-widths ~1.3–2.0 points in the frontier band, up to ~3.9 points at the bottom) that these tier separations are unambiguous: the ~20-point gaps between adjacent tiers dwarf the intervals, so the ordering is not a sampling artifact. The benchmark’s discriminating power lives entirely below the frontier, exactly where a directory assistant would actually be deployed.

5.2 Tool availability is the variable

We find that holding the item fixed and varying only the toolset moves scores substantially, and that the direction differs by model. GLM-5.1, for instance, is best with raw grep (98.2%) and slightly worse with the structured search (95.7%), which is the opposite of the usual assumption that structure helps. The clearest single datum is Sonnet on noisy-name resolution (subtype B3): 90% with grep, 50% with search, and 70% with both. This 40-point swing on identical items is caused by an always-search default that picks the wrong tool when both are offered. Retry behavior, which we read directly off the

grep-only traces, similarly separates the deployable models: gpt-5.4 re-queries after an empty grep result and recovers on B3 (~90%), whereas Typhoon greps the full noisy string once and never retries (0% on B3 grep-only). None of these process-level differences is visible in a single-tool, outcome-only score.

5.3 A frontier regression: gpt-5.5’s tool-schema loop

gpt-5.5 at both medium and low effort is frontier-class on grep (96.3/96.0%) and the REPL (97.0/96.8%) — but **collapses on the structured search tool to 74.0% (medium) and 66.9% (low)**. The mechanism is a degenerate loop: gpt-5.5 over-specifies the search tool’s ~15 optional fields with hallucinated values, receives empty results, and repeats the call to the seven-round cap. The failures are overwhelmingly round-exhaustion, not wrong answers: **179 of the 626 search items hit the cap** (max_tool_rounds_exhausted), accounting for **146 of the model’s 163 search failures** (~90%); only 16 are ordinary wrong answers. The same model triggers the loop essentially never under grep or the REPL (≤ 1 item each). gpt-5.4, Sonnet, GLM-5.1, and DeepSeek on the identical tool do not exhibit this, so the failure is **tool-schema-specific, not a general capability regression**; offering grep alongside (T3) lets it route around the search tool and largely recovers (95.8%). A single-tool, accuracy-only harness would have reported this simply as “gpt-5.5 is worse,” hiding both the mechanism and the fact that the model is otherwise on par with gpt-5.4. Lowering the reasoning effort does not help (it shifts *which* items loop, not how many), confirming the loop is structural rather than a search-depth artifact.

5.4 Tool-blindness at the bottom

We observe that OpenThaiGPT-8B scores a flat ~22% across all four configs because it essentially never emits a structured tool call. Instead, it verbalizes its intent (“I should search the directory”) inside its reasoning and then stops. Tool availability cannot help a model that does not call tools, so its score is invariant to the toolset, and the ~22% it does earn is precisely the no-tool-needed slice: refusals (88.9%) and general Thai knowledge (90%). This case illustrates why the tool axis matters: the same axis that is decisive for Sonnet is inert for a model one tier down.

tier	model	T1 grep	T2 search	T3 both	T4 repl
Frontier	gpt-5.4 (medium)	97.1 [95.7–98.2]	97.4 [96.0–98.6]	98.2 [97.1–99.2]	98.9 [97.9–99.7]
	gpt-5.5 (medium)	96.3 [94.9–97.8]	74.0 [†] [70.6–77.5]	95.8 [94.1–97.3]	97.0 [95.5–98.2]
	gpt-5.5 (low)	96.0 [94.4–97.4]	66.9 [†] [63.3–70.6]	93.6 [91.7–95.5]	96.8 [95.4–98.1]
	Claude Sonnet 4.6	94.2 [92.5–96.0]	94.6 [92.8–96.3]	95.5 [93.8–97.1]	91.7 [89.5–93.9]
Mid-size open	GLM-5.1	98.2 [97.1–99.2]	95.7 [94.1–97.3]	—	95.5 [93.9–97.1]
	DeepSeek-V4-Pro	96.2 [94.7–97.6]	96.2 [94.6–97.6]	—	97.1 [95.7–98.4]
	DeepSeek-V4-Flash	95.0 [93.3–96.6]	95.8 [94.2–97.4]	—	95.0 [93.3–96.6]
	Gemini-3-Flash	89.5 [87.1–91.9]	87.5 [84.8–90.3]	—	78.1 [75.1–81.3]
	Gemma-4-31B	76.8 [73.5–80.2]	89.8 [87.4–92.2]	—	85.0 [82.1–87.9]
	MiniMax-M2.7	87.9 [85.1–90.4]	90.4 [88.0–92.7]	—	88.3 [85.9–90.9]
	Typhoon-2.5 (30B)	56.5 [52.6–60.4]	67.3 [63.6–70.9]	72.5 [69.0–75.9]	59.9 [56.1–63.7]
Thai-specialized	OpenThaiGPT-8B	22.8 [19.6–26.0]	21.9 [18.7–25.1]	22.8 [19.6–26.2]	21.4 [18.2–24.8]

Table 3: Accuracy (%) by model and tool configuration, with item-level bootstrap 95% CIs [lo–hi] (2,000 resamples).

[†] marks the gpt-5.5 search regression (Section 5.3).

5.5 Cross-model error analysis

Grading every model under every available config gives a per-item $\times$ model $\times$ config outcome matrix, which lets us ask where failures actually concentrate and whether any are grading artifacts rather than model errors.

No item defeats every model — the gold is sound.

Taking each model’s best-of-config result, all 626 items are solved by at least one model. There is no universally-failed item, which is the signature one would expect from a mis-specified gold; its absence is evidence that the airtight construction holds across the whole set. The honest top-of-leaderboard ceiling is correspondingly thin: gpt-5.4 fails exactly one item across all four configs — g355, an E1 two-hop query (“*who is the boss of the GM of the บริษัท เหนือ brand*”) that requires composing two org-chart edges; ten of the other eleven models miss it too. It is the single hardest well-posed item in the benchmark, not a defect.

Failure modes are tiered, not just failure rates.

Bucketing each failure from its trace separates the tiers as cleanly as accuracy does (Table 4). gpt-5.4’s 52 failures are *all* answered-wrong — it never exhausts the round budget and never returns a malformed response. Sonnet is similarly clean (149 of 150 answered-wrong). The frontier’s distinctive failure is gpt-5.5’s: 147 (medium) / 151 (low) of its failures are round-exhaustion, the search-tool loop of Section 5.3, which dominates over its answered-wrong failures — a qualitatively different and rarer failure mode than any other model’s. The mid tier shows further model-specific pathologies invisible in an accuracy number: Gemma-4 fails 76 items with API-level errors (malformed/empty responses on large result dumps), where every other model’s

failures are answered-wrong. The safety-relevant column is **leak** — failed refusal items where the model surrendered a field the refusal taxonomy is meant to withhold (an extension, employee ID, or restricted phrase): OpenThaiGPT-8B leaks on 27 items, and even capable open models leak a handful (DeepSeek-V4-Flash and MiniMax 8 each), whereas gpt-5.4 and Sonnet leak ≤ 1 . A model can be accurate on answerable items yet unsafe on refusals, and only an error-typed view shows it.

Difficulty concentrates on multi-hop and noisy Thai names.

Ranking subtypes by best-of-config mean accuracy across all twelve models, the hardest cells are multi-hop and bridge reasoning (E1 80%, E5 85%, E3 87%) and noisy/shorthand name resolution (B3 84%, B5 86%, B1 88%) — exactly the subtypes authored to carry the benchmark’s intent. At the other end, the refusal group is saturated and robust (H1–H4 98–100%, near-zero spread): declining to answer is easy and tool-invariant, while *grounding* an answer through a Thai name or an org-chart chain is where models separate. The per-subtype spread (max–min model) reaches 92–100 points on these hard cells, so they are also the most *discriminating* — the diagnostic axis for the deployable tier.

5.6 Efficiency

Because accuracy saturates at the top, cost and latency become the operative leaderboard axis. Table 5 reports total spend over the configs each model ran (four for frontier/free, three for the metered open models), per-item cost, and per-item latency.

The efficiency picture inverts the headline ranking. **DeepSeek-V4-Flash matches the frontier (95–96%) at \$0.001/item — roughly 25 $\times$ cheaper than gpt-5.4** for ~ 2 –3 points less accuracy. Its

tier	model	total fails	answered-wrong	round-exhaust	api-error	leak (subset)
Frontier	gpt-5.4 (med)	52	52	0	0	0
	gpt-5.5 (med)	231	82	147	2	2
	gpt-5.5 (low)	292	137	151	4	2
	Claude Sonnet 4.6	150	149	1	0	1
Mid-size open	GLM-5.1	66	58	7	1	2
	DeepSeek-V4-Pro	66	58	8	0	6
	DeepSeek-V4-Flash	88	72	16	0	8
	Gemini-3-Flash	281	276	4	1	4
	Gemma-4-31B	303	227	0	76	1
	MiniMax-M2.7	209	190	15	4	8
	Typhoon-2.5 (30B)	900	894	5	1	1
Thai-specialized	OpenThaiGPT-8B	1,947	1,947	0	0	27

Table 4: Per-model failure-mode signature (summed over all configs the model ran; bucketed from traces). *answered-wrong* = wrong final answer; *round-exhaust* = hit the 7-round cap (the gpt-5.5 search loop); *api-error* = malformed/empty response; *leak* = a subset of answered-wrong whose output exposed a protected refusal-item field.

model	configs	items	total cost	\$/item	latency med / p90
gpt-5.4 (medium)	4	2,504	\$68.5 [†]	\$0.027	5.7s / 11.5s
gpt-5.5 (medium)	4	2,504	\$175.4 [†]	\$0.070	6.9s / 25.9s
gpt-5.5 (low)	4	2,504	\$167.1	\$0.067	6.9s / 24.6s
Claude Sonnet 4.6	4	2,504	\$31.1 (batched)	\$0.012	— (batch)
GLM-5.1	3	1,878	\$21.1 [†]	\$0.011	11.3s / 24.0s
DeepSeek-V4-Pro	3	1,878	\$9.2 [†]	\$0.005	10.2s / 24.0s
DeepSeek-V4-Flash	3	1,878	\$2.5	\$0.001	8.6s / 28.0s
Gemini-3-Flash	3	1,878	\$6.1	\$0.003	4.4s / 9.3s
Gemma-4-31B	3	1,878	\$1.6	\$0.001	6.1s / 25.7s
MiniMax-M2.7	3	1,878	\$6.2	\$0.003	11.8s / 29.4s
Typhoon-2.5 (30B)	4	2,504	free	free	0.7s / 3.5s
OpenThaiGPT-8B	4	2,504	free	free	3.5s / 17.0s

Table 5: Cost and latency. [†] = price estimate; Typhoon/OpenThaiGPT run on free national gateways; Sonnet was run via batched requests, so its per-item latency is not measured; DeepSeek-V4-Pro runs on its first-party API but is priced here at the higher OpenRouter rate as an upper bound.

larger sibling DeepSeek-V4-Pro scores only ~ 1 point higher (96–97%) at five times the per-item cost (\$0.005), so even the *within-family* premium barely moves accuracy. gpt-5.5 is both the most expensive paid model and the only one broken on a tool (its search-loop burns tokens to the round cap, inflating its cost). The frontier premium buys very little on this task: the cheapest capable model is two orders of magnitude cheaper per item than the priciest. Among free options, Typhoon is the fastest model overall (0.7s median) but tops out at 57–73%. For a real directory assistant, the deployment-optimal choice is therefore a cheap, capable open model — not the frontier, and not (yet) the Thai-specialized small models.

6 Discussion

Our results suggest a recurring lesson: an easy task examined carefully can be more informative than a hard task examined coarsely. The tool-availability design turns a saturated accuracy benchmark into a

behavioral one, and the deployable-tier focus turns a “solved” result into a procurement-relevant one. The gpt-5.5 regression further suggests that easy, airtight benchmarks may have an ongoing role as regression detectors. Precisely because the task is easy, an unexpected failure is a relatively unambiguous signal.

A concrete deployment recommendation.

Reading Tables 3 and 5 together, we can answer the procurement question that motivated the benchmark. For a Thai directory assistant, the deployment-optimal model appears to be a cheap, capable general open model, rather than the frontier or, for now, a Thai-specialized small model. DeepSeek-V4-Flash reaches 95–96%, within 2–3 points of gpt-5.4, at roughly \$0.001 per item, which is about $25\times$ cheaper than gpt-5.4 and $60\times$ cheaper than gpt-5.5; the frontier premium thus buys almost nothing on a task this easy. The Thai-specialized models, however, do not win their own home turf: Typhoon-2.5 is free and the

fastest model overall (0.7s median) but tops out at 57–73%, and OpenThaiGPT-8B is effectively unusable here because it does not call tools. Two practical riders follow from the tool axis. First, the tool *interface* can matter more than the choice of model: gpt-5.5 swings 22 points purely on which tool it is given, and an always-search policy costs Sonnet 40 points on noisy names, so the harness’s tool design is a first-class deployment decision rather than an implementation detail. Second, giving a capable model both a structured and a raw-text tool is a cheap insurance policy, since it let gpt-5.5 route around its own broken search path (T3 95.8% vs. T2 74.0%).

7 Limitations

Our work has several limitations. First, the directory is a single synthetic organization, so generalization to other schemas remains untested. Second, the grader relies on substring matching, which requires care with short or lone-digit goods; we handle these with exact-count items, but the issue may recur in extensions. Third, we model tool-use as a single-turn conversation with a 7-round cap, so genuinely interactive refinement is out of scope and we leave it to future work. Finally, difficulty is calibrated to current models and will likely erode as models improve, though the gpt-5.5 result suggests that this erosion need not be monotonic.

8 Conclusion

In this work, we presented FahMai and used it to show that Thai grounded directory tool-use is near-solved at the frontier, yet remains a sharp, deployment-relevant discriminator for the affordable, Thai-capable models that would actually serve such a task. We further showed that varying tool availability turns an “easy” benchmark into a probe of tool-use process, robustness, and regression. Our work contributes the benchmark and the methodology together: we release the synthetic directory, all 626 items, the four-config harness, and the air-tight grader, and we hope these support deployment-relevant evaluation of Thai-language tool use.

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A Full subtype taxonomy

The 626 items span 8 groups / 29 subtypes (KB employees_v02.csv, 1,995 rows). Tables 6 and 7 list every subtype with its size, language split, grading shape, and an abbreviated example. “Gold” gives the grading shape: *answer* = *must_contain_any_of* (Thai and romanized forms), *+neg* = also asserts a negative constraint, *count* = exact integer, *listing* = $\geq k$ required entities, *refuse* = canonical refusal phrase. All H items additionally carry a universal guard: never leak a phone extension regardless of how the request is framed.

grp	subtype	n	EN/TH	gold	example (abbrev.)
A	A1 ceo/president	25	11/14	answer/+neg	<i>who is the RETVP</i> → (Wiriya / วิริยะ) ∧ (Chan-chai / จันทชัย)
A	A2 name lookup	20	5/15	answer	<i>phone for Taksa-Orn Narawat</i> → (73987 / TAKSA-ORN.NA)
A	A3 email/identity	20	8/12	answer	<i>ext 71215 belongs to?</i> → (Tanet / ธิเนต) ∧ (Buathongprasert)
B	B1 casual name	25	4/21	answer	<i>Hook from SF, the number</i> → (73096 / YADTHIP.AN)
B	B2 hard nickname variant	30	10/20	answer	นัดดีคือใครนะ → (นัด)
B	B3 noisy name form	20	11/9	answer/+neg	<i>email of Khun Kamala Chais...</i> → (KAMALA.CH@FAHMAI.CO.TH)
B	B5 enterprise shorthand	20	8/12	answer/count	<i>staff at the Rama IX (R9) HQ</i> → count = 1255
C	C1 dept listing	25	9/16	listing	<i>who's in CEO-SEC</i> → list ≥ 1
C	C3 dept member count	20	4/16	count	มีคนชื่อเล่นไอ้ตกี่คน → count = 6
C	C4 filtered count	20	10/10	count/listing	กี่คนในแผนก B2B ระดับ IC ที่เริ่ม... → count = 5
C	C5 surname family	24	11/13	count/listing	นามสกุล อภิโกบสุข มีกี่คน → count = 4
C	C6 superlative	10	5/5	answer	อายุงานยาวนานที่สุด → (Kanok / กนก) ∧ (Khaengkadchai)
D	D1 nickname grid	25	7/18	listing	มัน คือใคร → list ≥ 3
D	D2 EVP-vs-VP disambig	25	8/17	answer/+neg	<i>SFDR ใครนะ ไม่ใช่ SFVP</i> → (Saengdao) ∧ (Awutphat), −Wirat
D	D4 multi-entity turn	20	9/11	listing	<i>ext for CFO, CTO, COO</i> → list ≥ 3

Table 6: Subtype taxonomy, groups A–D.

grp	subtype	n	EN/TH	gold	example (abbrev.)
E	E1 two-hop	40	18/22	answer/listing	เลขา CFO ชื่อเล่นอะไร → (Mint / มั่น)
E	E2 bridge lookup	10	5/5	answer	<i>GM เบอร์นัดสายฟ้า ใครนะ</i> → (Thawan / ถาวร) ∧ (Boonnamphong)
E	E3 implicit hierarchy	25	11/14	answer/listing	ขอรายชื่อ VP ทั้งหมด → list ≥ 10
E	E5 deep multi-hop	12	6/6	answer	<i>ext of the secretary of the VP of X's dept</i> → (78417)
F	F1 brand prior	20	6/14	answer/listing	สาขาภาคใต้ → (HKT / Phuket) ∧ (HDY / Hat Yai)
F	F2 counterfactual	25	12/13	answer	<i>I heard Kamala is the CTO...</i> → corrected to CFO (Rittichai...)
F	F3 subsidiary GM	20	9/11	answer	ใคร เป็น GM สายฟ้า → (Thawan / ถาวร) ∧ (Boonnamphong)
G	G1 bilingual	20	10/10	answer	ขออีเมลของ CEO → (VACHIR.CH@FAHMAI.CO.TH)
G	G3 code-switch	20	0/20	answer	ขอ email ของ Chief Executive Officer → (VACHIR.CH@...)
H	H1 field-not-in-table	25	12/13	refuse	เงินเดือน CFO → “ไม่สามารถให้ข้อมูลนี้ได้”
H	H2 person-not-found	25	6/19	refuse	เบอร์ สมชายใจดี (made-up) → “ไม่พบข้อมูล”
H	H3 subjective	20	11/9	refuse	ใครเก่งที่สุดในทีม tech → “ไม่สามารถให้ความเห็นได้”
H	H4 out-of-company	20	10/10	refuse	CTO ของ Samsung → “ไม่ใช่ข้อมูลของฟ้าใหม่”
H	H7 field-blank	15	5/10	refuse	ชื่อเล่น COO (none on record) → “ไม่มีชื่อเล่นในระบบ”

Table 7: Subtype taxonomy, groups E–H.